Predicting Vinyl Sales in the 21st Century

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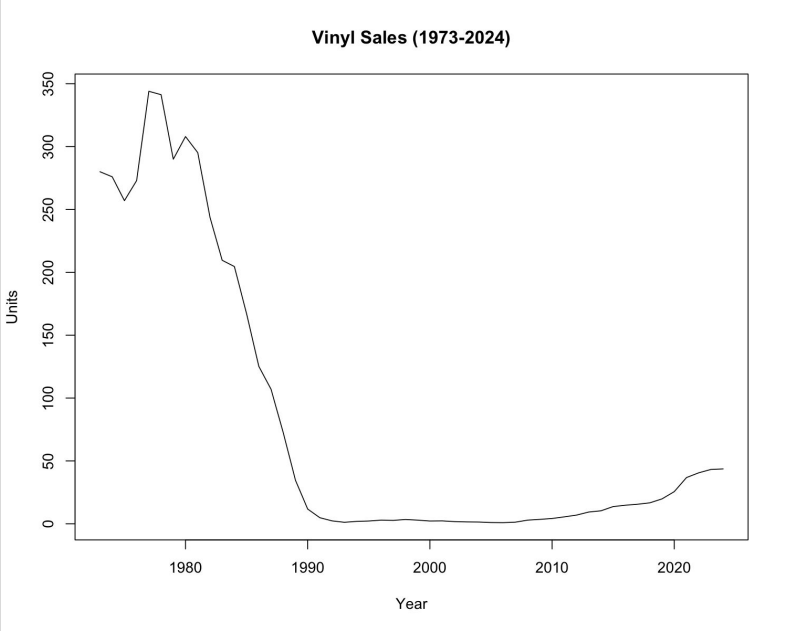
Math 443: Professor Piryatinska

Goal/Motivation

- With the proliferation of streaming, consumers are starting to once again buy physical media.
- This includes CDs, cassette tapes, and the most popular: Vinyl Records
- I noticed, from getting into them around 15 years ago, that it seems to be bigger than ever (in my lifetime)
- I would like to see if time series can come up with a model to predict the next 5-10 years of sales

The Data (Raw)

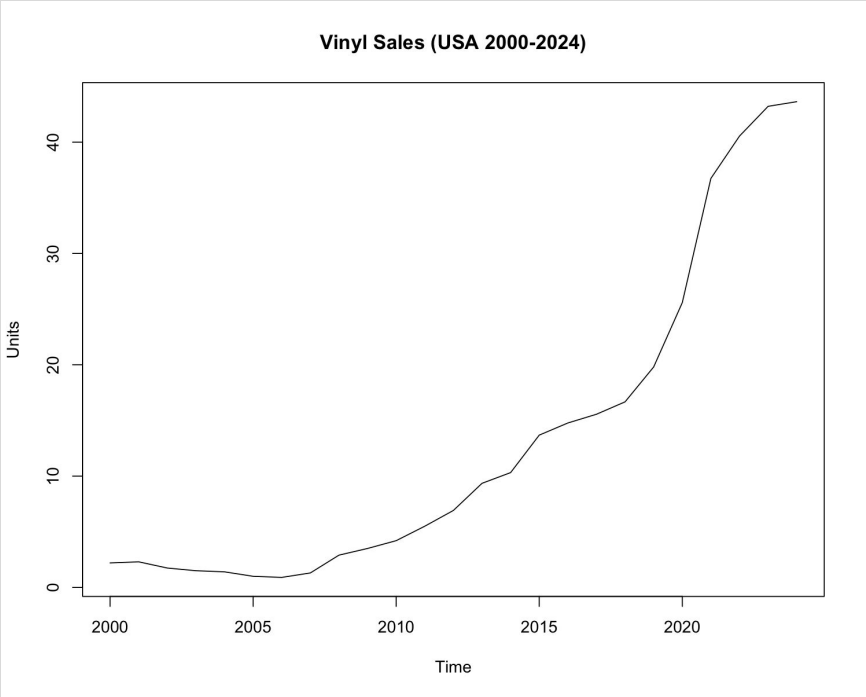
- I got the data from the public RIAA yearly sales reports. Here it is plotted ~1970-2024



- Clearly too large of a set, and enough factors have changed that i'd like to narrow down to 21st century

The Data (Reconfigured)

- Reconfigured data to have years 2000-2024.



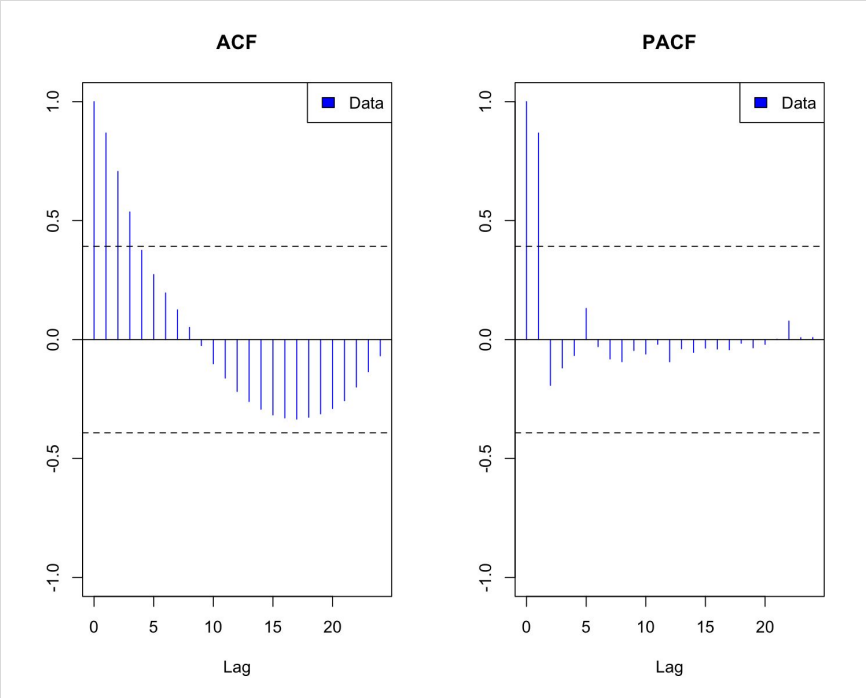
Minimum	Q1	Median	Mean	Q3	maximum
0.9	2.2	6.9	13	16.7	43.6

Table 1: Summary of the Time Series

- Appears to have an exponential trend, but can check a few trend models

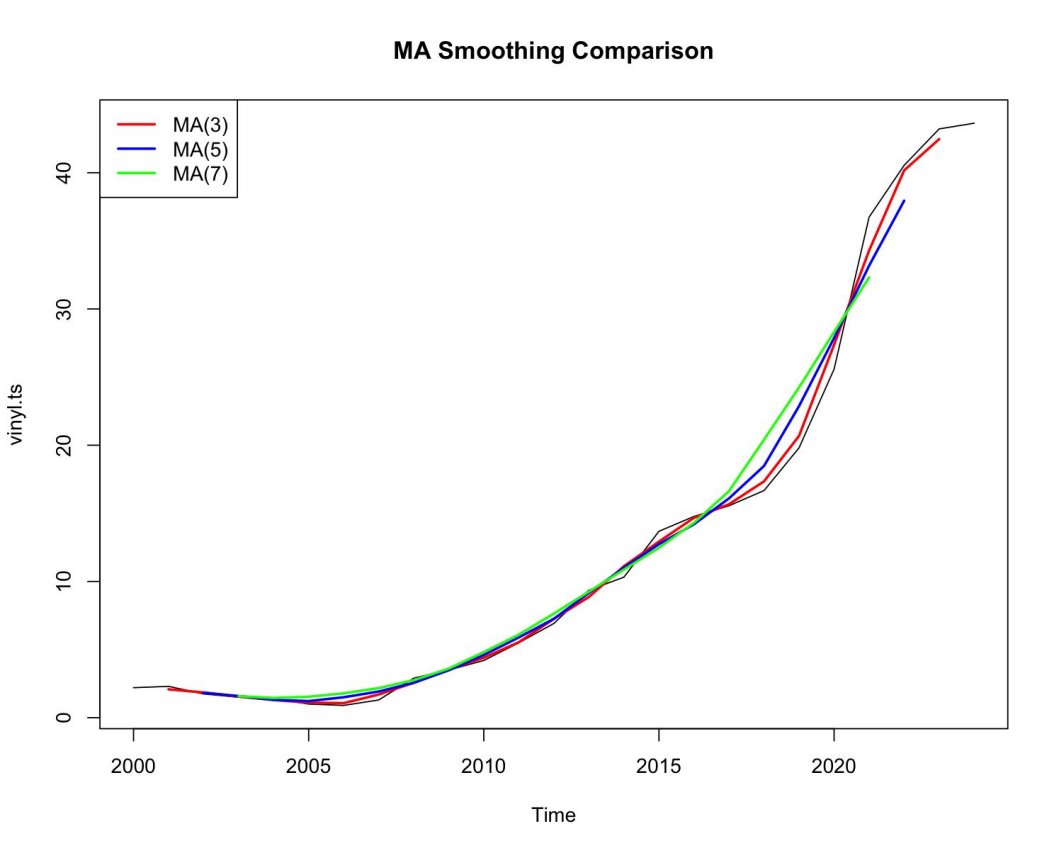
The Data (ACF/PACF)

- ACF/PACF for data can help narrow down models.



- ACF
  - Slow decay means there is a trend, so may need to do some differencing.
- PACF
  - Shows AR component 1 or 2

MA Smoothing



- Smooths the curve, but due to low number of indices (25), going too high (7) starts to eliminate data.

Stationary Testing

```
Augmented Dickey-Fuller Test
alternative: stationary

Type 1: no drift no trend
     lag   ADF p.value
[1,]   0 4.345   0.990
[2,]   1 0.802   0.866
[3,]   2 0.998   0.910
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 2.497   0.990
[2,]   1 0.211   0.965
[3,]   2 0.424   0.978
Type 3: with drift and trend
     lag    ADF p.value
[1,]   0 -0.738   0.956
[2,]   1 -1.670   0.688
[3,]   2 -1.230   0.864
----
Note: in fact, p.value = 0.01 means p.value <= 0.01
```

```
Augmented Dickey-Fuller Test
alternative: stationary

Type 1: no drift no trend
     lag   ADF p.value
[1,]   0 -1.86  0.0632
[2,]   1 -1.59  0.1031
[3,]   2 -1.69  0.0868
Type 2: with drift no trend
     lag   ADF p.value
[1,]   0 -2.35   0.206
[2,]   1 -2.19   0.262
[3,]   2 -2.44   0.173
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -2.82   0.251
[2,]   1 -2.78   0.268
[3,]   2 -4.39   0.010
----
Note: in fact, p.value = 0.01 means p.value <= 0.01
```

```
Augmented Dickey-Fuller Test
alternative: stationary

Type 1: no drift no trend
     lag    ADF p.value
[1,]   0 -1.146  0.2606
[2,]   1 -2.067  0.0405
[3,]   2  0.456  0.7664
Type 2: with drift no trend
     lag    ADF p.value
[1,]   0 -1.683   0.445
[2,]   1 -2.934   0.058
[3,]   2 -0.489   0.859
Type 3: with drift and trend
     lag   ADF p.value
[1,]   0 -1.95   0.576
[2,]   1 -5.48   0.010
[3,]   2 -2.68   0.304
----
Note: in fact, p.value = 0.01 means p.value <= 0.01
```

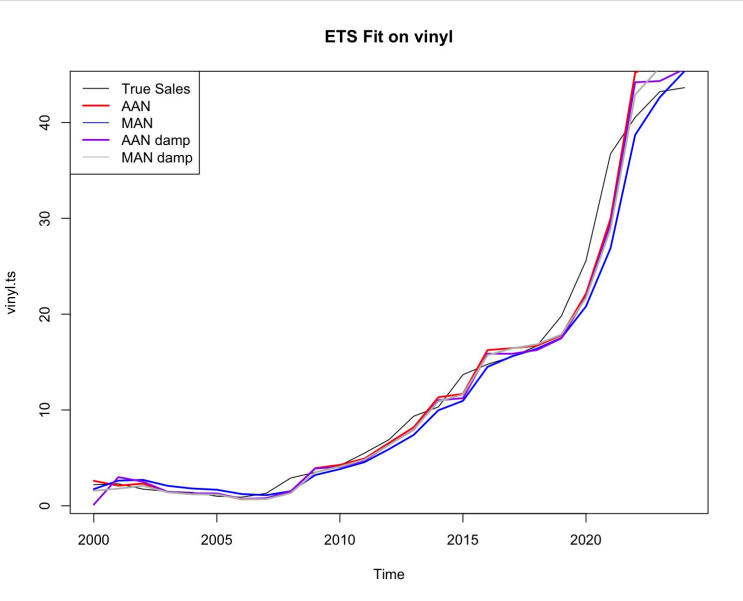
- Fails for series, but first and second difference is stationary!

Modeling (Exponential)

```
> test(residuals(ets.add))
Null hypothesis: Residuals are iid noise.
Test                        Distribution Statistic   p-value
Ljung-Box Q                Q ~ chisq(20)     11.12     0.943
McLeod-Li Q                Q ~ chisq(20)     24.47    0.2226
Turning points T     (T-15.3)/2 ~ N(0,1)        15    0.8696
Diff signs S         (S-12)/1.5 ~ N(0,1)        16    0.0066 *
Rank P             (P-150)/21.4 ~ N(0,1)       151    0.9627
> #adf.test(residuals(ets.add))
> test(residuals(ets.mult))
Null hypothesis: Residuals are iid noise.
Test                        Distribution Statistic   p-value
Ljung-Box Q                Q ~ chisq(20)     15.86    0.7251
McLeod-Li Q                Q ~ chisq(20)      3.31         1
Turning points T     (T-15.3)/2 ~ N(0,1)        12    0.1006
Diff signs S         (S-12)/1.5 ~ N(0,1)        13    0.4969
Rank P             (P-150)/21.4 ~ N(0,1)       170    0.3502
> #adf.test(residuals(ets.mult))
> test(residuals(ets.dampa))
Null hypothesis: Residuals are iid noise.
Test                        Distribution Statistic   p-value
Ljung-Box Q                Q ~ chisq(20)      9.32    0.9789
McLeod-Li Q                Q ~ chisq(20)     10.01     0.968
Turning points T     (T-15.3)/2 ~ N(0,1)        14    0.5114
Diff signs S         (S-12)/1.5 ~ N(0,1)        15    0.0415 *
Rank P             (P-150)/21.4 ~ N(0,1)       156    0.7793
> #adf.test(residuals(ets.dampa))
> test(residuals(ets.dampm))
Null hypothesis: Residuals are iid noise.
Test                        Distribution Statistic   p-value
Ljung-Box Q                Q ~ chisq(20)     14.86    0.7842
McLeod-Li Q                Q ~ chisq(20)      6.62    0.9977
Turning points T     (T-15.3)/2 ~ N(0,1)        14    0.5114
Diff signs S         (S-12)/1.5 ~ N(0,1)        15    0.0415 *
Rank P             (P-150)/21.4 ~ N(0,1)       120    0.1611
```

```
> ets.aic
     AAN      MAN   AAN(D)   MAN(D)
129.0967 117.8408 129.4551 126.1886
```

- The AAN, MAN, AANd, MANd all pass iid noise residuals (except diff signs test), though MAN is the best of these models (lowest AIC).



Modeling (ARIMA)

- Tried ARIMA(1,1,0), ARIMA(1,1,1) and ARIMA(2,1,0)

ARIMA 110	ARIMA 111	ARIMA 210	ARIMA 020
107.7108	109.6984	109.6930	105.0332

- All models residuals are good, AIC also lower than exponential

- Auto ARIMA returned an ARIMA(0,2,0)

- ARIMA(0,2,0) is the best model here!

Overall Model Comparison

AAN	MAN	AAN(D)	MAN(D)	ARIMA 110	ARIMA 111	ARIMA 210	ARIMA 020
129.0967	117.8408	129.4551	126.1886	107.7108	109.6984	109.6930	105.0332

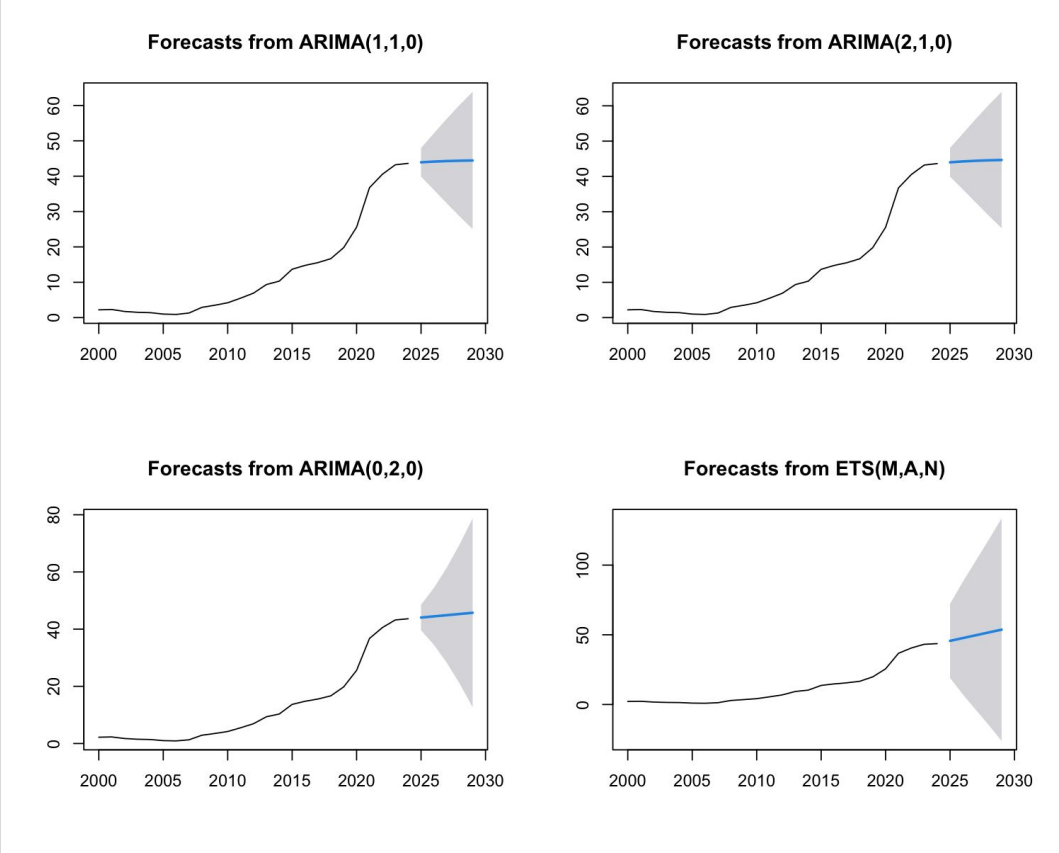
- Clearly all ARIMA models outperform exponential models.
- Surprisingly, the ARIMA(0,2,0) with only differencing is the best model. Also easy to interpret.
- The two best are ARIMA(1,1,0) and ARIMA(0,2,0)

Forecasting (5 years)

- Although the “best model” is (0,2,0), it has larger bounds when forecasting 5 years.
- Exponential is definitely the worst

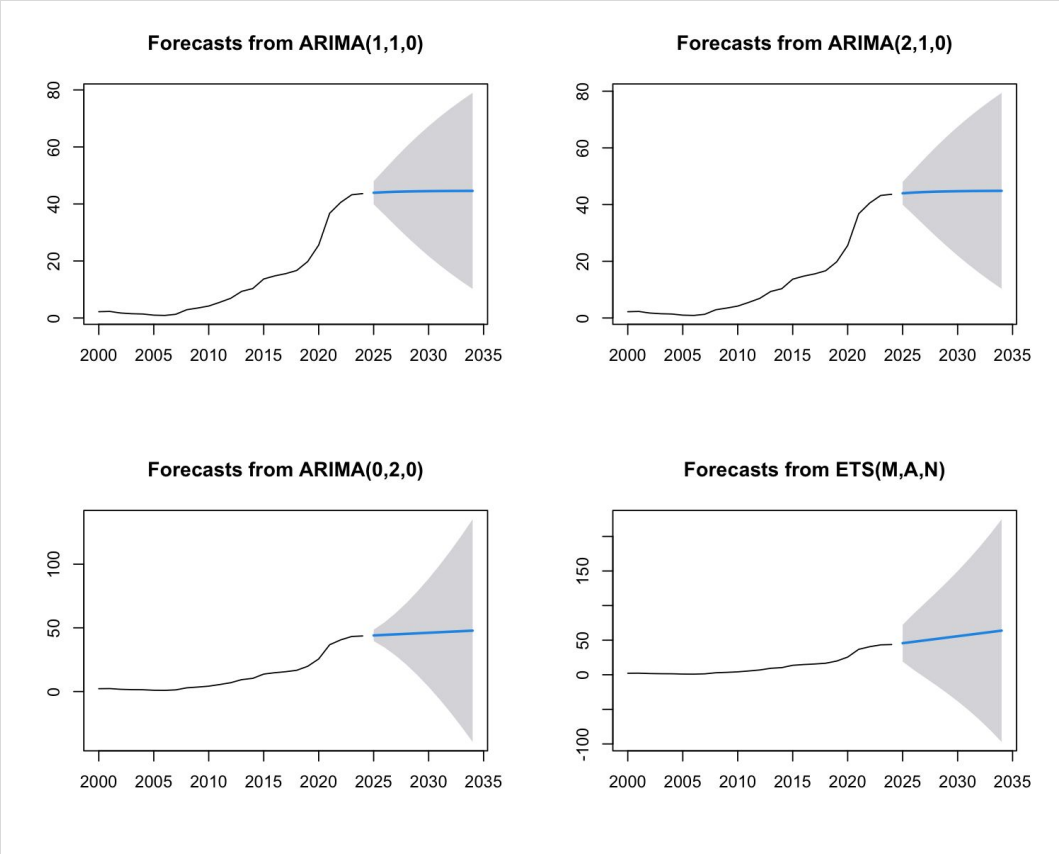
Model	Forecast	Lower	Upper
ARIMA(1,1,0)	44.45	25.04	63.87
ARIMA(2,1,0)	44.65	25.35	63.96
ARIMA(0,2,0)	45.72	12.68	78.75
ETS(M,A,N)	53.71	-26.13	133.55

Table 3: 2025 Forecasts



Forecasting (10 years)

- Similar to 5 years
- No model not great at forecasting this long.



Conclusions/Next Steps

- Overall, the models were alright at prediction, but the confidence intervals get too large over longer prediction periods. ARIMA(0,2,0) the best AIC, but ARIMA(1,1,0) and (2,1,0) have the better predictions/forecasts

- Some next steps
  - Get more granular data (monthly)
  - Compare to other forms of media (CD, streaming)
  - Investigate other patterns about vinyl, maybe production
  - Use more advanced models.